

**INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES**

# **INDUSTRIAL BOILER SYSTEM**

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

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Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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## 1. System Profiling & Overview

An industrial boiler generates steam or hot water by transferring heat from a fuel source (coal, furnace oil, natural gas, biomass, or waste heat) to water under pressure. Boiler reliability and efficiency depend critically on feed water quality, since scale and corrosion inside a pressurized, high-temperature vessel can cause tube failure, efficiency loss, and safety hazards.

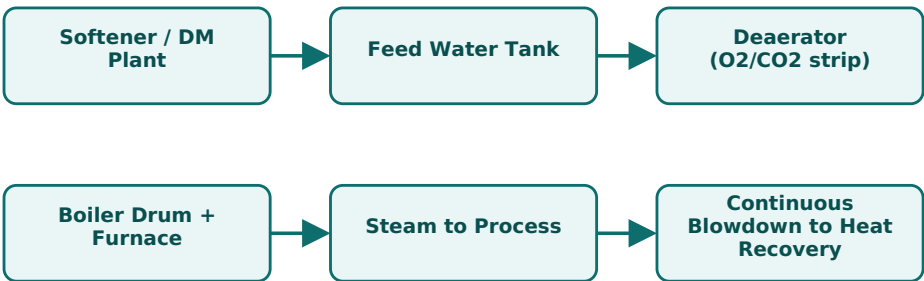


Fig. 1 — Boiler Feed Water & Steam Generation Path

### 1.1 Applications

- Process steam generation for chemical, textile, food, pharmaceutical and paper industries.
- Power generation (utility and captive power plants).
- Space heating and hot water supply for institutional/commercial buildings.
- Sterilization (autoclaves) in pharmaceutical and healthcare facilities.

### 1.2 Boiler Classification by Pressure

| Category                    | Typical Operating Pressure | Feed Water Quality Requirement                                                            |
|-----------------------------|----------------------------|-------------------------------------------------------------------------------------------|
| Low Pressure                | Up to 10 kg/cm2            | Softened water generally sufficient                                                       |
| Medium Pressure             | 10-40 kg/cm2               | DM water recommended, softened water with careful chemical dosing acceptable at lower end |
| High Pressure               | 40-100+ kg/cm2             | DM/Mixed bed water mandatory; strict silica and iron limits                               |
| Supercritical/Utility Scale | >170 kg/cm2                | Ultra-pure DM/condensate polished water, extremely tight limits on all impurities         |

## 2. Boiler Types & Selection Basis

| Type                                  | Description                                                                                   | Typical Application                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Fire Tube Boiler                      | Hot flue gases pass through tubes submerged in water; simple, robust                          | Low-to-medium pressure process steam (up to ~17-20 kg/cm2), small-medium industries |
| Water Tube Boiler                     | Water/steam flows inside tubes surrounded by hot gases; higher pressure/capacity capability   | Medium-to-high pressure, power plants and large process industries                  |
| Package Boiler                        | Factory-assembled, skid-mounted fire tube or water tube unit                                  | Quick deployment for standard capacity industrial needs                             |
| Waste Heat Recovery Boiler            | Uses waste heat from process/exhaust gases instead of primary fuel                            | Cement, steel, chemical plants with significant waste heat streams                  |
| Fluidized Bed Combustion (FBC) Boiler | Solid fuel (coal/biomass) combusted in a fluidized bed for efficient, low-emission combustion | Solid fuel-fired industrial and power boilers                                       |

## 3. Feed & Boiler Water Quality Design Basis

Feed water and boiler water limits are typically specified per IS 10392 (Indian Standard) or ASME/EPRI guidelines and vary by operating pressure – higher pressure demands progressively tighter limits since higher temperature/pressure increases scaling and corrosion sensitivity.

| Parameter                              | Low Pressure Boiler                 | High Pressure Boiler                                     |
|----------------------------------------|-------------------------------------|----------------------------------------------------------|
| Feed Water Hardness                    | ≤1-2 mg/L as CaCO3                  | Nil (DM water)                                           |
| Feed Water Dissolved Oxygen            | ≤0.1-0.5 mg/L (post deaeration)     | ≤0.007 mg/L (post deaeration + oxygen scavenger)         |
| Feed Water pH                          | 8.5-9.5                             | 8.8-9.2                                                  |
| Boiler Water TDS                       | 2000-3500 mg/L (pressure dependent) | ≤50-150 mg/L (very high pressure)                        |
| Boiler Water Silica                    | ≤150-300 mg/L (low pressure)        | ≤2-20 mg/L (pressure dependent, per steam purity limit)  |
| Boiler Water Alkalinity (P-alkalinity) | Controlled per TDS/silica ratio     | Very tightly controlled to prevent caustic embrittlement |

#### 4. Pretreatment & Internal Water Treatment

| Stage                                | Purpose                                                                       | Typical Equipment                                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| External Softening/DM                | Remove hardness/dissolved solids before feed tank                             | Softener or DM plant (see respective manuals)                                                                                                                           |
| Deaerator                            | Remove dissolved oxygen and CO2 from feed water using steam stripping         | Spray-cum-tray or packed-column deaerator, typically operating at 5-10°C above feed inlet saturation                                                                    |
| Chemical Dosing (Internal Treatment) | Control residual hardness precipitation, scale, and corrosion inside boiler   | Phosphate/polymer dosing for scale control, oxygen scavenger (sodium sulfite/hydrazine alternatives) for corrosion control, amine dosing for condensate line protection |
| Condensate Polishing (high pressure) | Remove corrosion products/contamination from returned condensate before reuse | Mixed bed condensate polisher                                                                                                                                           |

#### 5. Blowdown, Deaeration & Steam Purity

- **Continuous Blowdown:** Controlled bleed-off of concentrated boiler water to maintain TDS within design limit, ideally with heat recovery (flash tank + heat exchanger) to minimize energy loss.
- **Intermittent (Bottom) Blowdown:** Periodic manual/automatic purge from the mud drum to remove settled sludge/solids, typically once or twice per shift.
- **Deaeration:** Mechanical + chemical removal of dissolved oxygen/CO2 is essential since these gases are the primary cause of pitting corrosion in boiler and condensate return lines.
- **Steam Purity:** Carryover of boiler water solids into steam (priming/foaming) can damage turbines and process equipment; controlled via TDS limits, proper drum internals (cyclone separators, steam scrubbers), and correct water level control.

## 6. Boiler Water Chemistry Deep-Dive

### 6.1 Internal Phosphate/Polymer Treatment Chemistry

Coordinated phosphate treatment:  $3\text{Ca}^{2+} + 2\text{PO}_4^{3-} \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s}) \downarrow$  (soft, non-adherent sludge, removed by blowdown, rather than hard adherent scale)

Internal phosphate treatment deliberately precipitates any residual hardness that slips past external softening/DM as a soft, mobile sludge that is removed by bottom blowdown – rather than allowing it to deposit as hard, adherent, insulating scale directly on tube surfaces. The Na:PO<sub>4</sub> ratio is controlled (coordinated phosphate programs keep free NaOH low) specifically to avoid the caustic embrittlement mechanism described in 6.3.

### 6.2 Oxygen Scavenging Chemistry

Sodium sulfite:  $2\text{Na}_2\text{SO}_3 + \text{O}_2 \rightarrow 2\text{Na}_2\text{SO}_4$

Hydrazine (high pressure, thermally decomposes, leaves no dissolved solids):  $\text{N}_2\text{H}_4 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{N}_2$

Sodium sulfite is preferred for low/medium pressure boilers since it is simple to dose and test, but it adds to boiler water TDS (as sodium sulfate) and decomposes to corrosive SO<sub>2</sub> species above ~275°C, limiting its use to lower pressure. Hydrazine (or modern non-hazardous alternatives such as carbohydrazide/erythorbate) leaves no dissolved solids behind and is used in high-pressure/utility boilers where TDS must be minimized, but requires careful handling due to toxicity concerns with hydrazine specifically.

### 6.3 Caustic Embrittlement / Stress Corrosion Cracking

Concentration mechanism: free NaOH concentrates in crevices/stressed rivet joints via localized boiling (steam blanketing), reaching much higher local causticity than the bulk boiler water average

Caustic embrittlement is a localized phenomenon – the bulk boiler water alkalinity may be well within limits, but at a stressed joint under a steam blanket, water can concentrate to a much higher local NaOH concentration, driving intergranular stress corrosion cracking. This is why alkalinity limits in Section 3 are set conservatively relative to steam pressure/temperature rather than to the bulk corrosivity of the boiler water alone, and why riveted/welded older boilers are especially monitored for this failure mode.

### 6.4 Steam Purity & Carryover Chemistry

Two distinct carryover mechanisms exist: mechanical carryover (fine water droplets physically entrained in steam, controlled by drum internals and water level) and vaporous carryover (silica specifically has a small but real volatility in saturated steam that increases sharply with pressure, per the boiler water silica:steam silica partition ratio). This is why boiler water silica limits (Section 3) tighten so much faster with rising pressure than TDS limits do – at very high pressure, silica volatilization becomes the dominant purity concern even with zero mechanical carryover.

## 7. Common Problems & Engineering Solutions

| Problem                                         | Root Cause                                                                                   | Engineering Solution                                                                         |
|-------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Scale formation on tubes/drum                   | Hardness in feed water, inadequate phosphate/polymer dosing                                  | Verify feed water softening/DM quality, correct internal chemical dosing per TDS             |
| Pitting corrosion in boiler/condensate lines    | Dissolved oxygen ingress, inadequate deaerator performance                                   | Check deaerator vent rate and temperature, verify oxygen scavenger residual                  |
| Caustic embrittlement/stress corrosion cracking | High free caustic alkalinity concentrating at stressed joints/rivets                         | Control alkalinity per steam pressure guidelines, add coordinated phosphate treatment        |
| Carryover/priming (wet steam)                   | High TDS, high water level, sudden load swings, oil contamination                            | Increase blowdown, correct water level control, check for oil ingress from condensate return |
| Low boiler efficiency/high fuel consumption     | Scale on heat transfer surfaces, poor combustion, blowdown losses without heat recovery      | Descale tubes, tune burner/combustion air ratio, install blowdown heat recovery              |
| Tube failure/bulging                            | Localized overheating from scale, low water level incident, DM water lacking water treatment | Root cause metallurgical investigation, review water treatment program                       |

|                               |                                                       |                                                                            |
|-------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------|
| Frequent safety valve lifting | Pressure control malfunction, burner modulation fault | Inspect pressure control loop, service safety valve per statutory schedule |
|-------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------|

## 8. Equipment Service Life

| Equipment                   | Typical Service Life                            | Key Life-limiting Factors                                                    |
|-----------------------------|-------------------------------------------------|------------------------------------------------------------------------------|
| Boiler Shell/Pressure Parts | 25-35 years (with proper water treatment)       | Scale/corrosion-induced tube failure, metal fatigue, water treatment quality |
| Boiler Tubes                | 10-20 years (varies greatly with water quality) | Scaling, corrosion, overheating incidents                                    |
| Burner/Combustion System    | 15-20 years                                     | Nozzle wear, refractory degradation, fuel quality                            |
| Deaerator Internals         | 15-20 years                                     | Corrosion, tray/spray nozzle erosion                                         |
| Feed Water Pumps            | 8-10 years (seals/bearings 2-3 yrs)             | Cavitation, high-temperature seal wear                                       |
| Safety Valves               | 10-15 years (annual testing mandatory)          | Seat wear, spring fatigue                                                    |
| Refractory Lining           | 3-7 years                                       | Thermal cycling, fuel ash/slag attack                                        |
| Economizer/Air Preheater    | 15-20 years                                     | Flue gas corrosion (acid dew point), fouling                                 |

## 9. Preventive Service & Maintenance Schedule

| Frequency                   | Activity                                                                                                                                                                                     |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daily                       | Test boiler water TDS/alkalinity/phosphate residual; check feed water hardness; log blowdown; verify deaerator operating temperature                                                         |
| Weekly                      | Test oxygen scavenger residual; inspect burner flame condition; check safety valve for leaks (do not test-lift routinely)                                                                    |
| Monthly                     | Full boiler/feed water lab analysis; inspect sight glass/level gauge; combustion efficiency (flue gas O2/CO) check                                                                           |
| Quarterly                   | Internal inspection via manhole (where accessible without shutdown); soot blower function check; economizer cleaning if fouled                                                               |
| Annual (Statutory Shutdown) | Complete internal/external inspection per statutory boiler inspector requirement, hydraulic test, safety valve setting verification, tube thickness (UT) survey, chemical cleaning if scaled |

## 10. Troubleshooting Guide

| Symptom                                   | Probable Cause                                                     | Corrective Action                                                               |
|-------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Boiler water TDS rising rapidly           | Insufficient blowdown, feed water quality degradation              | Increase blowdown rate, check upstream softener/DM performance                  |
| Low steam pressure at rated load          | Fouled tubes, burner underfiring, feed water starvation            | Inspect for scale, check burner turndown/fuel supply, verify feed pump capacity |
| Water hammer in steam lines               | Condensate accumulation, inadequate steam trap drainage            | Inspect/service steam traps, verify line slope and drain point spacing          |
| High stack temperature (efficiency loss)  | Fouled heat transfer surfaces (soot/scale), excess air             | Clean fireside/waterside surfaces, tune combustion air ratio                    |
| Frequent low-water cutoff trips           | Feed pump malfunction, level control fault, blowdown valve leaking | Inspect feed pump and control valve, check level gauge glass for blockage       |
| Corrosion pitting found during inspection | Inadequate deaeration or oxygen scavenger dosing history           | Review historical water treatment logs, reinforce deaerator/scavenger control   |

## 11. Project Execution & Engineering Workflow

| Phase             | Key Deliverables / Activities                                                             |
|-------------------|-------------------------------------------------------------------------------------------|
| Load & Fuel Study | Steam demand profile, fuel availability/selection, boiler type and capacity finalization. |

|                                          |                                                                                                                                            |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Basic Engineering                        | PFD, feed water treatment scheme selection, layout concept, capex-opex estimate.                                                           |
| Detailed Engineering                     | P&ID, GA drawings, structural design for boiler house, electrical SLD, control/interlock philosophy (per statutory IBR/ASME requirements). |
| Statutory Approvals                      | Design approval from Chief Inspector of Boilers (India, IBR) or applicable jurisdictional boiler authority before fabrication.             |
| Procurement                              | Boiler pressure parts (IBR/ASME-code fabrication), burner, feed water treatment package, instrumentation.                                  |
| Civil & Structural                       | Boiler house foundation, chimney/stack, fuel storage/handling area.                                                                        |
| Erection & MEI                           | Pressure part erection under statutory inspector supervision, piping, cabling, control panel wiring.                                       |
| Statutory Hydraulic Test & Certification | Hydro test at 1.5x design pressure, inspection and certification by boiler inspector before commissioning.                                 |
| Commissioning                            | Light-up, safety valve setting/testing, combustion tuning, water treatment system start-up.                                                |
| Performance Guarantee Test Run           | Verify guaranteed efficiency, steam output and quality at rated load over a sustained test.                                                |
| Handover & O&M                           | O&M manual, statutory certificate, operator training (certified boiler operator mandatory), spares philosophy.                             |

## 12. Statutory Compliance & Performance Guarantee

- Design, fabrication, installation and operation of boilers in India are governed by the Boilers Act and Indian Boiler Regulations (IBR); internationally, ASME Boiler and Pressure Vessel Code (Section I) is widely followed.
- Mandatory periodic statutory inspection (typically annual) by a certified boiler inspector, including internal/external inspection and hydraulic testing, is required to renew the operating certificate.
- Certified boiler operators (per statutory competency requirements) are mandatory for boiler operation above notified capacity/pressure thresholds.
- Performance guarantee typically covers thermal efficiency (%), fuel consumption per unit steam output, and steam purity (TDS/silica carryover), demonstrated over a sustained load test witnessed by client/consultant.

Technical note: figures in this manual are general design guidelines. Final boiler selection, water treatment scheme and operating limits must be established per statutory code requirements and actual fuel/feed water data.